2.65 Numerical Methods: Euler's Step by Step Method

Worked Solutions

1 a)

$$
\begin{aligned}
&x_0 = 2, \quad y_0 = 1 \\
&y_1 = 1 + 0.5f(2) = 1 + 0.5 \times 0.0183156889\ldots = 1.009157819\ldots \\
&x_1 = 2 + 0.5 = 2.5 \\
&y_2 = y_1 + 0.5f(x_1) = 1.009157819\ldots + 0.5 \times 0.001930454136\ldots \\
&\quad = 1.01012304\ldots \\
&x_2 = 2.5 + 0.5 = 3 \\
&\therefore y = 1.010 \text{ (3 DP)}
\end{aligned}
$$

1 b)

$$
\begin{aligned}
&x_0 = 2, \quad y_0 = 1 \\
&y_1 = 1 + 0.25f(2) = 1 + 0.25 \times 0.018315688\ldots = 1.00457890972\ldots \\
&x_1 = 2 + 0.25 = 2.25 \\
&y_2 = y_1 + 0.25f(x_1) = 1.00457890972\ldots + 0.25 \times 0.00632971542749\ldots \\
&\quad = 1.006161613\ldots \\
&x_2 = 2.25 + 0.25 = 2.5 \\
&y_3 = y_2 + 0.25f(x_2) = 1.006161613\ldots + 0.25 \times 0.001930454\ldots \\
&\quad = 1.00664395211\ldots \\
&x_3 = 2.5 + 0.25 = 2.75 \\
&y_4 = y_3 + 0.25f(x_3) = 1.00664395211\ldots + 0.25 \times 0.00051957\ldots \\
&\quad = 1.00677384578\ldots \\
&x_4 = 2.75 + 0.25 = 3 \\
&\therefore y = 1.007 \text{ (3 DP)}
\end{aligned}
$$

2 a)

$$
\begin{aligned}
&x_0 = 0, \quad y_0 = 1 \\
&y_1 = y_0 + hf(x_0, y_0) = 1 + 0.2f(0, 1) = 1 + 0.2 \times (-1) = 0.8 \\
&x_1 = 0 + 0.2 = 0.2 \\
&y_2 = y_1 + hf(x_1, y_1) = 0.8 + 0.2f(0.2, 0.8) \\
&\quad = 0.8 + 0.2 \times (-0.76) = 0.648 \\
&x_2 = 0.2 + 0.2 = 0.4 \\
&\therefore y = 0.648 \text{ (3 DP)}
\end{aligned}
$$

2 b)

$$
\begin{aligned}
&x_0 = 0, \quad y_0 = 1 \\
&y_1 = y_0 + hf(x_0, y_0) = 1 + hf(0, 1) = 1 + 0.1 \times (-1) = 0.9 \\
&x_1 = 0 + 0.1 = 0.1 \\
&y_2 = y_1 + hf(x_1, y_1) = 0.9 + 0.1f(0.1, 0.9) = 0.9 + 0.1 \times (-0.89) = 0.811 \\
&x_2 = 0.1 + 0.1 = 0.2 \\
&y_3 = y_2 + hf(x_2, y_2) = 0.811 + 0.1f(0.2, 0.811) = 0.811 + 0.1 \times (-0.771) = 0.7339 \\
&x_3 = 0.2 + 0.1 = 0.3 \\
&y_4 = y_3 + hf(x_3, y_3) = 0.7339 + 0.1f(0.3, 0.7339) = 0.7339 + 0.1 \times (-0.6439) = 0.66951 \\
&x_4 = 0.3 + 0.1 = 0.4 \\
&\therefore y = 0.670 \text{ (3 DP)}
\end{aligned}
$$

3 a)

$$
\begin{aligned}
&x_0 = 0, \quad y_0 = 2 \\
&y_1 = y_0 + hf(x_0, y_0) = 2 + 0.2f(0, 2) = 2 + 0.2 \times (-2) = 1.6 \\
&x_1 = 0 + 0.2 = 0.2 \\
&y_2 = y_1 + hf(x_1, y_1) = 1.6 + 0.2f(0.2, 1.6) \\
&\quad = 1.6 + 0.2 \times (-1.4013306692\ldots) = 1.3197338662\ldots \\
&x_2 = 0.2 + 0.2 = 0.4 \\
&\therefore y = 1.320 \text{ (3 DP)}
\end{aligned}
$$

3 b)

$$
\begin{aligned}
&x_0 = 0, \quad y_0 = 2 \\
&y_1 = 2 + 0.1f(0, 2) = 2 + 0.1 \times (-2) = 1.8 \\
&x_1 = 0 + 0.1 = 0.1 \\
&y_2 = 1.8 + 0.1f(0.1, 1.8) = 1.8 + 0.1 \times (-1.7001665834\ldots) = 1.6299833417\ldots \\
&x_2 = 0.1 + 0.1 = 0.2 \\
&y_3 = 1.6299833417\ldots + 0.1f(0.2, 1.6299833417\ldots) \\
&\quad = 1.6299833417\ldots + 0.1 \times (-1.4313140109\ldots) = 1.486851941\ldots \\
&x_3 = 0.2 + 0.1 = 0.3 \\
&y_4 = 1.486851941\ldots + 0.1f(0.3, 1.486851941\ldots) \\
&\quad = 1.486851941\ldots + 0.1 \times (-1.1913317339) = 1.3677187672\ldots \\
&x_4 = 0.3 + 0.1 = 0.4 \\
&\therefore y = 1.368 \text{ (3 DP)}
\end{aligned}
$$

4)

$$
\begin{aligned}
&x_0 = 2, \quad y_0 = -1, \quad h = 0.5 \\
&y_1 = y_0 + hf(x_0, y_0) = -1 + 0.5f(2, -1) = -1 + 0.5 \times (-3) = -2.5 \\
&x_1 = 2 + 0.5 = 2.5 \\
&y_2 = y_1 + hf(x_1, y_1) = -2.5 + 0.5f(2.5, -2.5) = -2.5 + 0.5 \times 1 = -2 \\
&x_2 = 2.5 + 0.5 = 3 \\
&\therefore y = -2
\end{aligned}
$$

5 a)

$$
\begin{aligned}
&\frac{dy}{dx} = \frac{e^{\sqrt{x}}}{2x}, \quad x_0 = 4, \quad y_0 = 1 \\
&y_1 = y_0 + hf(x_0) = 1 + 0.5f(4) = 1 + 0.5 \times 0.9236320124\ldots = 1.46186006\ldots \\
&x_1 = 4 + 0.5 = 4.5 \\
&y_2 = y_1 + hf(x_1) = 1.46186006\ldots + 0.5f(4.5) \\
&\quad = 1.46186006\ldots + 0.5 \times 0.9269049685\ldots = 1.92526849\ldots \\
&x_2 = 4.5 + 0.5 = 5 \\
&\therefore y = 1.925 \text{ (3 DP)}
\end{aligned}
$$

5 b)

$$
\begin{aligned}
&x_0 = 0, \quad y_0 = 2 \\
&y_1 = 2 + 0.5f(0, 2) = 2 + 0.5 \times 0.3465735903\ldots = 2.1732867951\ldots \\
&x_1 = 0 + 0.5 = 0.5 \\
&y_2 = y_1 + 0.5f(x_1, y_1) = 2.1732867951\ldots + 0.5f(0.5, 2.1732867951\ldots) \\
&\quad = 2.1732867951\ldots + 0.5 \times 0.4072747327\ldots = 2.3769241615\ldots \\
&x_2 = 0.5 + 0.5 = 1 \\
&y_3 = y_2 + 0.5f(1, y_2) = 2.3769241615\ldots + 0.5f(1, 2.3769241615\ldots) \\
&\quad = 2.3769241615\ldots + 0.5 \times 0.5119916334\ldots = 2.6329199781\ldots \\
&x_3 = 1 + 0.5 = 1.5 \\
&\therefore y = 2.633 \text{ (3 DP)}
\end{aligned}
$$

6 a)

$$
\begin{aligned}
&x_0 = 0, \quad y_0 = 1 \\
&y_1 = y_0 + hf(x_0) = 1 + 0.2f(0) = 1 + 0.2 \times 2 = 1.4 \\
&x_1 = 0 + 0.2 = 0.2 \\
&y_2 = y_1 + hf(x_1) = 1.4 + 0.2f(0.2) = 1.4 + 0.2 \times 2e^{0.4}\ldots = 1.996729879\ldots \\
&x_2 = 0.2 + 0.2 = 0.4
\end{aligned}
$$

$$
\begin{aligned}
y_3 &= y_2 + hf(x_2) = 1.996729879\ldots + 0.2f(0.4) \\
&= 1.996729879\ldots + 0.2 \times 2e^{0.8}\ldots = 2.88694625\ldots \\
x_3 &= 0.4 + 0.2 = 0.6 \\
\therefore y &= 2.887 \text{ (3 DP)}
\end{aligned}
$$

6 b)

Separating and integrating gives $y = e^{2x} + c$. Since $y(0) = 1$, $1 = e^0 + c$, so $c = 0$ and $y = e^{2x}$. When $x = 0.6$, $y = e^{1.2}$.

6 c)

The relative error is

$$\frac{2.887 - e^{1.2}}{e^{1.2}} = -0.130.$$

6 d)

The relative error is negative, which means the approximate value is less than the exact value. Therefore, the Euler approximation underestimates the true value of $y(0.6)$.

7 a)

$$
\begin{aligned}
x_0 &= 1, \quad y_0 = 4 \\
y_1 &= y_0 + hf(x_0, y_0) = 4 + 0.4f(1, 4) = 4 + 0.4 \times (-0.9609308626\ldots) = 3.6156276550\ldots \\
x_1 &= 1 + 0.4 = 1.4 \\
y_2 &= y_1 + hf(x_1, y_1) = 3.6156276550\ldots + 0.4f(1.4, 3.6156276550\ldots) \\
&= 3.6156276550\ldots + 0.4 \times (-1.2692009154\ldots) = 3.1079472888\ldots \\
x_2 &= 1.4 + 0.4 = 1.8 \\
y_3 &= y_2 + hf(x_2, y_2) = 3.1079472888\ldots + 0.4f(1.8, 3.1079472888\ldots) \\
&= 3.1079472888\ldots + 0.4 \times (-1.1983021443\ldots) = 2.6286264311\ldots \\
x_3 &= 1.8 + 0.4 = 2.2 \\
\therefore y &= 2.629 \text{ (3 DP)}
\end{aligned}
$$

7 b)

Not very reliable as Euler's step by step method assumes the gradient is constant over each step, which is not true here since $\cos y$ changes rapidly.

7 c)

Increase the number of steps (use a smaller value for h).

8)

$$
\begin{aligned}
t_0 &= 0, \quad C_0 = 0 \\
C_1 &= C_0 + hf(t_0, C_0) = 0 + 0.5f(0, 0) = 0 + 0.5 \times 5 = 2.5 \\
t_1 &= 0 + 0.5 = 0.5
\end{aligned}
$$

$$C_2 = C_1 + hf(t_1, C_1) = 2.5 + 0.5f(0.5, 2.5) = 2.5 + 0.5 \times 4.55 = 4.775$$
$$t_2 = 0.5 + 0.5 = 1$$
$$\therefore C \approx 4.775 \text{ mg}$$

9)

$$x_0 = 0, \quad y_0 = 12000$$
$$y_1 = y_0 + hf(x_0, y_0) = 12000 + f(0, 12000) = 12060$$
$$x_1 = 0 + 1 = 1$$
$$y_2 = y_1 + hf(x_1, y_1) = 12060 + f(1, 12060) = 12220.3$$
$$x_2 = 1 + 1 = 2$$
$$y_3 = y_2 + hf(x_2, y_2) = 12220.3 + f(2, 12220.3)$$
$$= 12220.3 + 234.3065808\ldots = 12454.6065808\ldots$$
$$x_3 = 2 + 1 = 3$$
$$\therefore y = 12455 \text{ people}$$

10)

$$t_0 = 5, \quad T_0 = 50$$
$$T_1 = T_0 + hf(t_0, T_0) = 50 + 1f(5, 50) = 50 + 5 \times (-1.5) = 48.5$$
$$t_1 = 5 + 1 = 6$$
$$T_2 = T_1 + hf(t_1, T_1) = 48.5 + 1f(6, 48.5) = 48.5 + 1 \times (-1.225) = 47.275$$
$$t_2 = 6 + 1 = 7$$
$$T_3 = T_2 + hf(t_2, T_2) = 47.275 + 1f(7, 47.275) = 47.275 + 1 \times (-0.963750) = 46.31125$$
$$t_3 = 7 + 1 = 8$$
$$\therefore T = 46.3^\circ\text{C (3 SF)}$$

11)

$$t_0 = 2, \quad B_0 = 970, \quad h = 1$$
$$B_1 = B_0 + hf(t_0, B_0) = 970 + 1f(2, 970) = 970 + 12.401511529\ldots = 982.401511529\ldots$$
$$t_1 = 2 + 1 = 3$$
$$B_2 = y_1 + hf(t_1, B_1) = 982.401511529\ldots + 1f(3, 982.401511529\ldots)$$
$$= 12.401511529\ldots + 10.177701124\ldots = 992.579212653\ldots$$
$$t_2 = 3 + 1 = 4$$
$$\therefore B = £992.58$$

12 a)

$$x_0 = 0, \quad y_0 = 1, \quad h = 0.1$$

$$
\begin{aligned}
y_1 &= y_0 + hf(x_0) = 1 + 0.1f(0) = 1 + 0.1 \times 0 = 1 \\
x_1 &= 0 + 0.1 = 0.1 \\
y_2 &= y_1 + hf(x_1) = 1 + 0.1f(0.1) = 1 + 0.1 \times 0.0998334167\ldots = 1.009983342\ldots \\
x_2 &= 0.1 + 0.1 = 0.2 \\
y_3 &= y_2 + hf(x_2) = 1.009983342\ldots + 0.1f(0.2) \\
&= 1.009983342\ldots + 0.1 \times 0.1986693308\ldots = 1.029850275\ldots \\
x_3 &= 0.2 + 0.1 = 0.3 \\
\therefore y &= 1.030 \text{ (3 DP)}
\end{aligned}
$$

12 b)

Since $\dfrac{dy}{dx} = \sin x$, then $y = -\cos x + c$. So y is concave, so the tangents lie below the curve, so each step gives a value that is slightly too small, so it is an under-estimate. Alternatively, integrating gives $y = -\cos x + c$. At $(0, 1)$, $1 = -\cos 0 + c$, so $c = 2$ and $y = -\cos x + 2$. At $x = 0.3$, $y = -\cos 0.3 + 2 = 1.044664\ldots > 1.30$, so this is an under-estimate.

13)

$$
\begin{aligned}
x_0 &= 3, \quad y_0 = a \\
y_1 &= y_0 + hf(x_0, y_0) = a + 0.2f(3, a) \\
a + 0.2 \times \frac{6 - a}{3a} &= 4.450 \\
3a^2 + 1.2 - 0.2a &= 13.35a \\
3a^2 - 13.55a + 1.2 &= 0 \\
a &= 0.09036897\ldots \quad \text{or} \quad a = 4.426298678\ldots \\
&\text{But } a > 2, \text{ so } a = 4.426
\end{aligned}
$$

14 a)

$$
\begin{aligned}
x_0 &= 1, \quad y_0 = 0.8 \\
y_1 &= y_0 + \frac{1}{k} f(x_0, y_0) = 0.8 + \frac{1}{k} f(1, 0.8) \\
&= 0.8 + \frac{1}{k}\sqrt{1.6 + 1} \\
0.8 + \frac{1}{k}\sqrt{2.64} &= 1.1250 \\
\frac{1}{k} &= \frac{1.1250 - 0.8}{\sqrt{2.64}} = 0.2000236728\ldots \\
\therefore \text{Since } k \text{ is an integer, } \frac{1}{k} &= \frac{1}{5}
\end{aligned}
$$

14 b)

$$
y_1 = 0.8 + \frac{1}{k} f(1, 0.8) = 0.8 + 0.2\sqrt{2.64} = 1.124961536\ldots
$$

$$
\begin{aligned}
x_1 &= 1 + 0.2 = 1.2 \\
y_2 &= y_1 + \frac{1}{k} f(x_1, y_1) = 1.124961536\ldots + 0.2 f(1.2, 1.124961536\ldots) \\
&= 1.124961536\ldots + 0.2 \times 1.914559599\ldots = 1.507873456\ldots \\
x_2 &= 1.2 + 0.2 = 1.4 \\
\therefore (x_2, y_2) &= (1.4, 1.508)
\end{aligned}
$$

15 a)

At each point (x_n, y_n), the derivative $\dfrac{dy}{dx} = x_n + y_n$ gives the gradient of the tangent line. The formula moves h units horizontally and then $h\times$ gradient vertically to obtain the next point (x_{n+1}, y_{n+1}), which is the end of a straight-line segment tangent to the curve at the previous point.

15 b)

$$
\begin{aligned}
x_0 &= 0, \quad y_0 = 1 \\
y_1 &= y_0 + h f(x_0, y_0) = 1 + 0.5 f(0, 1) = 1 + 0.5 \times 1 = 1.5 \\
x_1 &= 0 + 0.5 = 0.5 \\
y_2 &= y_1 + h f(x_1, y_1) = 1.5 + 0.5 f(0.5, 1.5) \\
&= 0.5 + 0.5 \times 2 = 2.5 \\
x_2 &= 0.5 + 0.5 = 1 \\
\therefore y &= 2.5
\end{aligned}
$$

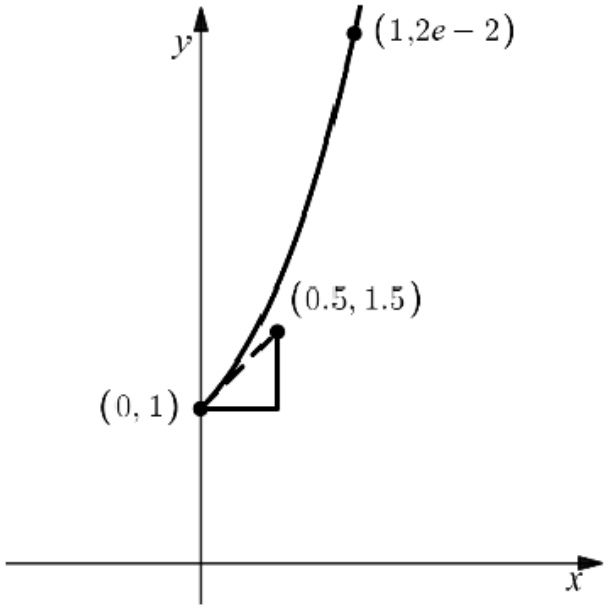
15 c)

$\dfrac{dy}{dx} - y = x$ has integrating factor $e^{\int -1\,dx} = e^{-x}$.

$$
\begin{aligned}
e^{-x}\frac{dy}{dx} - e^{-x} y &= x e^{-x} \\
y e^{-x} &= \int x e^{-x}\,dx \\
u = x, \quad \frac{dv}{dx} &= e^{-x} \\
\frac{du}{dx} = 1, \quad v &= -e^{-x} \\
\therefore \int x e^{-x}\,dx &= -x e^{-x} + \int e^{-x}\,dx = -x e^{-x} - e^{-x} (+c) \\
y e^{-x} &= -x e^{-x} - e^{-x} + c \\
y &= -x - 1 + c e^{x} \\
y(0) = 1: \quad 1 &= 0 - 1 + c \quad \therefore c = 2 \\
y &= 2e^{x} - x - 1 \\
\therefore y(1) &= 2e - 2
\end{aligned}
$$

15 d)

Sketch from the source worked-solutions PDF; it shows the curve $y = 2e^x - x - 1$, the tangent at $(0, 1)$, and the first Euler step to $(0.5, 1.5)$.



16 a)

Let the rate be $\frac{dy}{dt} = kt$. Since $24 = 4k$, $k = 6$, so $\frac{dy}{dt} = 6t$.

16 b)

$$
\begin{aligned}
t_0 &= 1, \quad y_0 = 1.5 \\
y_1 &= y_0 + hf(t_0) = 1.5 + 0.25f(1) = 1.5 + 0.25 \times 6 = 3 \\
t_1 &= 1 + 0.25 = 1.25 \\
y_2 &= y_1 + hf(t_1) = 3 + 0.25f(1.25) = 3 + 0.25 \times 7.5 = 4.875 \\
t_2 &= 1.25 + 0.25 = 1.5 \\
\therefore y &= 4.88 \text{ GB (2 DP)}
\end{aligned}
$$

17)

$$
\begin{aligned}
\frac{dN}{dt} &= 0.08N - 500e^{-t} \\
t_0 &= 1, \quad N_0 = 4700 \\
N_1 &= N_0 + hf(t_0, N_0) = 4700 + 0.5f(1, 4700) \\
&= 4700 + 0.5 \times 192.060279414\ldots = 4796.030139707\ldots \\
t_1 &= 1 + 0.5 = 1.5
\end{aligned}
$$

$$
\begin{aligned}
N_2 &= N_1 + hf(t_1, N_1) = 4796.030139707\ldots + 0.5f(1.5, 4796.030139707\ldots) \\
&= 4796.030139707\ldots + 0.5 \times 272.117331102\ldots \\
&= 4932.088805258\ldots \\
t_2 &= 1.5 + 0.5 = 2 \\
\therefore N &= 4932 \text{ views}
\end{aligned}
$$

18)

$$
\begin{aligned}
\frac{dW}{dt} &= 0.5 - 2\sin^2\left(\frac{\pi}{12}t\right) - 0.01W \\
t_0 &= 0, \quad W_0 = 100 \\
W_1 &= W_0 + hf(t_0, W_0) = 1000 + 3f(0, 1000) = 1000 + 3 \times (-0.5) = 98.5 \\
t_1 &= 0 + 3 = 3 \\
W_2 &= W_1 + hf(t_1, W_1) = 98.5 + 3f(3, 98.5) = 98.5 + 3 \times (-1.485) = 94.045 \\
t_2 &= 3 + 3 = 6 \\
W_3 &= W_2 + hf(t_2, W_2) = 94.045 + 3f(6, 94.045) = 94.045 + 3 \times (-2.44045) = 86.72365 \\
t_3 &= 3 + 6 = 9 \\
\therefore W &= 86.7 \text{ litres (3 SF)}
\end{aligned}
$$